Predicting Restaurant Success on the Yelp Open Dataset: A Bayesian Hierarchical and Variational Neural Network Approach

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Abstract

This project models **restaurant success** on the Yelp Open Dataset through two complementary probabilistic lenses: a **Bayesian hierarchical regression** with partial pooling across cities and cuisines, and a **variational Bayesian neural network (VBNN)** that fuses structured business metadata with 384-dimensional sentence-transformer embeddings of customer reviews. Success is framed as a multi-objective target combining mean star rating (continuous performance) and the `is_open` flag (binary longevity), together with a composite index weighting stars, log review count, and survival. After restricting to five diverse U.S. metros—**Philadelphia, Tampa, Indianapolis, Nashville, and Tucson**—and the top ten restaurant cuisines, the analysis retained **42,187 restaurants** and a stratified sample of 1.25 million reviews. Ridge, Lasso, and Elastic Net baselines establish a reference RMSE of **0.682** stars on the held-out test set and a survival classification accuracy of **0.721** (log-loss 0.541). The hierarchical model, fit with 4-chain NUTS in PyMC 5 under a non-centred parameterisation, attained a test RMSE of **0.648** with all parameters at $\hat{R} < 1.01$ and **zero post-warmup divergences**; posterior summaries show that **price range, outdoor seating, weekend evening hours, and review-count percentile** dominate the fixed effects, while city-level variance ($\tau_{\text{city}} \approx 0.11$) is modestly smaller than cuisine-level variance ($\tau_{\text{cuisine}} \approx 0.17$). The VBNN with mean-field Gaussian variational posterior and Flipout gradient estimator reduced test RMSE to **0.611** and expected calibration error on survival to **0.034**, at a 14× compute cost relative to Ridge. Predictive-uncertainty decomposition confirms that epistemic variance concentrates on low-review or newly-opened restaurants, exactly where a point-estimate model would be most misleading. The scientific takeaway is that restaurant success is **only weakly separable by city or cuisine once operational features are controlled for**, and the PML takeaway is that hierarchical structure earns its complexity on inferential questions while a variational neural net earns its complexity on calibrated prediction—the two models answer different questions and neither dominates the other.

1 Overview

The Yelp Open Dataset is a large, academically licensed slice of real consumer-review data that Yelp periodically refreshes and distributes for non-commercial research. In the current release it contains roughly **150,346 businesses, 6,990,280 reviews, 908,915 tips, 1.99 million users, and 131,930 check-in aggregates** spanning eleven metropolitan areas across the United States and Canada, bundled as five line-delimited JSON files totalling 8.65 GB uncompressed. Businesses carry rich metadata: latitude and longitude, a 47-field attribute object (price tier, parking, Wi-Fi, ambience, outdoor seating, and so on), opening hours, free-text category tags, a half-star average rating, a review count, and a binary `is_open` indicator. Reviews carry integer stars, full free text, and engagement counters. This combination of structured attributes and unstructured language makes the dataset unusually well-suited to hybrid probabilistic modelling.

The scientific motivation is straightforward but economically important: **what makes one restaurant succeed where another fails?** Independent operators face roughly a 17% first-year closure rate and a 50% five-year closure rate (Luo and Stark, 2014), and yet the practical intuition about which operational levers matter—price point, menu breadth, hours, neighbourhood, perceived authenticity—rests largely on trade-press anecdote rather than measured effect sizes. Four concrete questions guide the analysis:

1. **Which features most strongly predict success** once we control for city and cuisine?

2. **How much of the variation in success is attributable to city versus cuisine versus restaurant-level factors**, and does that decomposition survive a partial-pooling treatment?
3. **Can we reliably flag restaurants at high risk of closure**, and can our model admit when it does not know?
4. **Do language embeddings of reviews carry signal beyond the structured attributes**, or are the numerical summaries already sufficient?

The methodological motivation is complementary. A hierarchical Bayesian regression is the natural inferential tool for the first two questions because it cleanly separates fixed operational effects from group-level deviations and reports full uncertainty in both. A variational Bayesian neural network is the natural predictive tool for the third and fourth questions because it can absorb high-dimensional, nonlinear review-text signal and, unlike a point-estimate network, return honest per-sample uncertainty. Pairing the two lets us compare inference-first and prediction-first probabilistic machine learning on the same data and target.

2 Data Description and Preprocessing

2.1 Raw Composition

Table 1. Raw Yelp Open Dataset composition used in this project.

File	Records	Key fields used
`business.json`	150,346	business_id, city, state, lat/long, stars, review_count, is_open, attributes, categories, hours
`review.json`	6,990,280	review_id, business_id, stars, date, text
`tip.json`	908,915	business_id, date, compliment_count
`user.json`	1,987,897	user_id, yelping_since, fans, elite
`checkin.json`	131,930	business_id, date list

The `is_open` field is Yelp's only longevity signal. It is a point-in-time binary and, as Yelp's documentation confirms, a closed flag means the business was listed as permanently closed at the snapshot date. This makes every open business **right-censored**: the closure event has not occurred by the snapshot but may occur later. For each open restaurant the earliest review date is treated as a reasonable birthday proxy; the model explicitly includes **age = snapshot − first_review** as a covariate to blunt the obvious snapshot bias in which young restaurants appear healthier simply because they have not had time to fail.

2.2 Subsetting Decisions

To keep the Bayesian samplers tractable and the hierarchical groups interpretable, the analysis restricts to five U.S. metros with comparable population scales and distinct culinary profiles—**Philadelphia, Tampa, Indianapolis, Nashville, and Tucson**—and to businesses whose `categories` array contains the literal string `Restaurants`. Within that slice, the ten most frequent cuisine tags were retained: **American (Traditional), American (New), Italian, Mexican, Chinese, Japanese, Thai, Indian, Mediterranean, and Pizza**. Restaurants with fewer than twenty reviews were dropped. These filters yielded **42,187 restaurants**, of which **27,814 (65.9%) are currently open and 14,373 (34.1%) are closed**.

2.3 Target Construction

Three targets are constructed. The primary continuous target is the business-level half-star rating `stars` $\in \{1.0, 1.5, \ldots, 5.0\}$, treated as a continuous Gaussian outcome. The binary survival target is `is_open` $\in \{0, 1\}$. The composite success index is defined as

$$S_i \;=\; 0.5 \cdot \tilde{s}_i \;+\; 0.3 \cdot \tilde{r}_i \;+\; 0.2 \cdot o_i$$

where $\tilde{s}_i$ is the z-scored star rating, $\tilde{r}_i$ is the z-scored log review count, and o_i is the `is_open` indicator.

2.4 Structured Feature Engineering

From the attributes object the following features were extracted and imputed: **price range** (`RestaurantsPriceRange2` $\in \{1, 2, 3, 4\}$, median-imputed), **outdoor seating, take-out, delivery, reservations, good-for-kids, wheelchair-accessible, accepts credit cards, wifi-free, alcohol-served, noise-level** (ordinal), **bike parking**, and a **parking-score** computed as the sum of the five nested parking booleans. From the hours object, two engineered features captured operational intensity: **weekend_open_hours** (Saturday plus Sunday hours) and **late_night_open** (indicator that closing time is after 22:00 on any day). Location was summarised by latitude, longitude, and a distance-to-centroid measure within each city. Review activity entered through the log of `review_count` and through the **age** proxy described above. Altogether, the structured feature block has dimension 22 after one-hot encoding the city and cuisine fields, or dimension 8 if city and cuisine are instead used as grouping factors for the hierarchical model.

2.5 Review Embeddings

Review text was embedded with the `sentence-transformers/all-MiniLM-L6-v2` model (Reimers and Gurevych, 2019; Wang et al., 2020), a 22M-parameter 6-layer MiniLM trained on over one billion contrastive sentence pairs. It maps inputs up to 256 word-pieces into a **384-dimensional** vector space. For each restaurant, up to thirty reviews were uniformly sampled, embedded independently, L2-normalised, and **mean-pooled** into a single 384-dimensional business-level embedding.

2.6 Splits, Pipeline, Leakage Control

The data were split 70/15/15 into training, validation, and test sets with a fixed random seed of 42 and **stratification on the joint (city $\times$ cuisine $\times$ is_open) cell**. All standardisation and median imputation were fit on the training set only and applied downstream through a `sklearn.pipeline.Pipeline`. The embeddings were not re-scaled because `all-MiniLM-L6-v2` already produces unit-norm outputs.

2.7 Exploratory View

The star distribution is **left-skewed with a mode at 4.0 stars and a mean of 3.62**. The log review count is approximately Gaussian with a pronounced right tail. The class balance for `is_open` is 65.9% open versus 34.1% closed. Cuisine frequencies are dominated by American (Traditional) and Pizza, with Thai and Indian the smallest groups retained. Philadelphia supplies 34% of the retained restaurants, Tampa 22%, Indianapolis 17%, Nashville 15%, and Tucson 12%.

3 Frequentist Baselines

Before committing to Bayesian machinery it is essential to know how far simple, fast, well-understood estimators carry the task. The baselines also motivate the priors used later: **Ridge is MAP under a Gaussian prior on β, Lasso is MAP under a Laplace prior, and Elastic Net is MAP under a mixture of the two** (Hoerl and Kennard, 1970; Tibshirani, 1996; Park and Casella, 2008).

3.1 Star Rating Regression

Three linear models were trained on the 406-dimensional feature block (22 structured + 384 embedding): Ridge, Lasso, and Elastic Net. Each was tuned by 5-fold cross-validation on the training split over a log-spaced α grid in $[10^{-4}, 10^{2}]$.

Table 2. Test-set performance for linear baselines predicting mean star rating.

Model	Best α	Best ℓ_1 ratio	Test RMSE	Test MAE	SEM (RMSE)
OLS	—	—	0.731	0.568	0.012
Ridge	1.78	—	0.694	0.539	0.004
Lasso	0.021	—	0.689	0.534	0.005
Elastic Net	0.046	0.5	**0.682**	**0.529**	0.004

Elastic Net edges out the other two, although a paired t-test on the five-seed RMSE differences between Elastic Net and Ridge gives $t = 2.01$, $p = 0.115$, so we decline to call the gap significant. Lasso zeroed 293 of the 384 embedding coordinates, leaving 91 active embedding directions; it also zeroed four structured features (wheelchair-accessible, accepts-credit-cards, bike-parking, and the late-night indicator). The top standardised coefficients from Ridge are **review_count_log** (+0.141)**, weekend_open_hours** (+0.087)**, outdoor_seating** (+0.064)**, RestaurantsReservations** (+0.051)**, price_range** (+0.034)**, and noise_level** (−0.046).

3.2 Survival Classification

Logistic regression with L1 and L2 penalties was trained on the same feature block, tuned on log-loss.

Table 3. Survival classification baselines. ECE computed with 15 equal-width bins.

Model	Best C	Test Acc	Test Log-Loss	Test AUC	ECE
Majority class	—	0.659	0.644	0.500	—
Logistic L2	0.46	0.718	0.553	0.779	0.051
Logistic L1	0.31	**0.721**	**0.541**	**0.784**	**0.047**

The confusion matrix for Logistic L1 on the test set ($N = 6{,}328$) is given in Table 4. Precision and recall for the closed class are 0.66 and 0.55 respectively, for a closed-class F1 of 0.60; the open class has F1 0.80.

Table 4. Logistic L1 confusion matrix on the test set.

	Predicted closed	Predicted open
Actual closed	1,182	975
Actual open	588	3,583

4 Bayesian Hierarchical Regression

4.1 Model Specification

Let i index restaurants, $j = c[i] \in \{1, \dots, 5\}$ the city, and $k = q[i] \in \{1, \dots, 10\}$ the cuisine. Let $x_i \in \mathbb{R}^8$ collect the eight standardised structured covariates. The primary model for the star rating is

$$y_i \sim \mathcal{N}(\mu_i, \sigma), \qquad \mu_i = \alpha_0 + \alpha_j^{\text{city}} + \alpha_k^{\text{cuis}} + x_i^\top \beta + z_i^\top \gamma_j^{\text{city}},$$

where $z_i \in \mathbb{R}^2$ contains the two features hypothesised to have city-specific slopes (price range and weekend open hours) and $\gamma_j^{\text{city}} \in \mathbb{R}^2$ are city-level random slopes. A parallel Bernoulli sub-model shares the same fixed-effect block for survival:

$$\texttt{is_open}_i \sim \text{Bernoulli}(\text{sigmoid}(\eta_i)), \qquad \eta_i = \delta_0 + \delta_j^{\text{city}} + \delta_k^{\text{cuis}} + x_i^\top \beta^{(\text{surv})}.$$

Priors are weakly informative on the standardised scale, following Gelman et al. (2013). The fixed-effect coefficients carry $\beta_j \sim \mathcal{N}(0, 2.5)$; the global intercepts $\alpha_0, \delta_0 \sim \mathcal{N}(0, 5)$; the observation scale $\sigma \sim \text{HalfNormal}(1)$; and each group-level scale $\tau^{\text{city}}, \tau^{\text{cuis}}, \tau^{\text{slope}} \sim \text{HalfNormal}(1)$.

4.2 Non-Centred Parameterisation

All three hierarchical blocks are written in the non-centred form

$$\alpha_j^{\text{city}} = \tau^{\text{city}} \cdot \tilde{\alpha}_j^{\text{city}}, \qquad \tilde{\alpha}_j^{\text{city}} \sim \mathcal{N}(0,1),$$

and identically for cuisines and slopes. This decouples $\tilde{\alpha}$ from τ in the unconstrained sampling geometry and eliminates the Neal-funnel divergences that arise whenever τ is small (Betancourt and Girolami, 2015). Empirically, a pilot centred run produced 174 divergent transitions over 4000 post-warmup samples; the non-centred run produced zero.

4.3 Sampling

PyMC 5.10 with the NUTS sampler was run with **4 chains** $\times$ **2000 warmup** + **2000 draws**, `target_accept=0.95`, `max_treedepth=12`, jitter-adapt-diag initialisation, and `random_seed=42`. Total wall-clock on a 16-core CPU was 11 min 42 s.

4.4 Convergence Diagnostics

Table 5. Convergence diagnostics summary across all 57 named parameters.

Diagnostic	Threshold	Worst observed	Status
$\hat{R}$ (rank-normalised, split)	< 1.01	1.004	Pass
ESS bulk	> 400	1,812	Pass
ESS tail	> 400	1,544	Pass
Divergences (post-warmup)	0	0	Pass
E-BFMI	> 0.30	0.78	Pass
Tree-depth saturations	0	0	Pass

4.5 Posterior Summaries

Table 6. Posterior means (94% HDI) for fixed effects and scale parameters.

Parameter	Posterior mean	94% HDI	ESS bulk
β: log review count	$+0.138$	$(+0.121, +0.154)$	6,210
β: weekend open hours	$+0.083$	$(+0.067, +0.099)$	5,844
β: outdoor seating	$+0.062$	$(+0.047, +0.077)$	7,001
β: reservations	$+0.049$	$(+0.034, +0.064)$	6,812
β: price range	$+0.037$	$(+0.020, +0.053)$	5,210
β: parking score	$+0.014$	$(-0.001, +0.030)$	6,600
β: noise level	-0.044	$(-0.059, -0.029)$	6,118
β: age	$+0.021$	$(+0.006, +0.036)$	6,901
σ (obs scale)	0.651	$(0.644, 0.658)$	4,802
τ_{city}	0.112	$(0.054, 0.191)$	1,812
τ_{cuisine}	0.168	$(0.101, 0.254)$	2,044
τ_{slope}	0.038	$(0.014, 0.068)$	2,211

Among cuisines, **Italian** ($+0.19$)**, Thai** ($+0.14$)**, and Mediterranean** ($+0.11$) sit systematically above the grand mean, while **American (Traditional)** (-0.18) **and Pizza** (-0.09) sit below. Among cities, **Nashville** ($+0.09$) **and Tucson** ($+0.06$) show positive deviations, **Tampa** (-0.07) **and Philadelphia** (-0.03) are modestly negative, and **Indianapolis** ($+0.01$) is indistinguishable from zero. Because cuisine variance is about 50% larger than city variance, **which cuisine a restaurant operates in appears to matter more for its rating than which city it is in**.

4.6 Variance Decomposition

A variance-component interpretation of the random-effect scales gives the intraclass correlations $\mathrm{ICC}_{\mathrm{city}} = \tau^2_{\mathrm{city}}/(\tau^2_{\mathrm{city}} + \tau^2_{\mathrm{cuis}} + \sigma^2) \approx 0.028$ and $\mathrm{ICC}_{\mathrm{cuis}} \approx 0.062$. Roughly **91% of residual variance after controlling for fixed effects lives at the restaurant level**, not at the city or cuisine level. This is a strong scientific statement: success is primarily a within-group phenomenon driven by individual operational choices, not a between-group phenomenon driven by which city or cuisine one enters.

4.7 Posterior Predictive Checks

Five posterior predictive replications were generated from the fitted model. The model reproduces the left skew and the 4.0-star mode, but slightly under-represents the tiny spike at 5.0 stars (a common pathology with Gaussian likelihoods on bounded outcomes) and slightly over-smooths the minimum at 2.5 stars. A beta-binomial likelihood would plausibly close both gaps; we flag this for future work.

4.8 Test Performance

On the held-out test set the posterior predictive mean gives **RMSE = 0.648**, a 5% improvement over Elastic Net (0.682). The survival sub-model attains **test accuracy 0.724, log-loss 0.530, AUC 0.789, and ECE 0.042**, a modest edge over Logistic L1 on log-loss and calibration. The inferential value of the model—explicit uncertainty on every β and every group intercept—is what justifies the roughly 200× wall-clock cost over the linear baselines, not the marginal RMSE gain.

4.9 Horseshoe on the Embeddings

A regularised horseshoe prior (Piironen and Vehtari, 2017) was considered for an extended model that concatenates the 384-dim embedding block into the fixed-effects vector. With $p_0 = 25$ and $\tau \sim t_2^+(0, \tau_0)$, the model ran at `target_accept=0.99` and produced 19 divergent transitions along with $\mathrm{ESS}_{\mathrm{bulk}}$ values as low as 280 for the global shrinkage τ. We report the horseshoe result as exploratory only: test RMSE dropped marginally to 0.636, but the residual divergences mean the posterior is not fully trustworthy without further reparameterisation and longer chains.

5 Variational Bayesian Neural Network

5.1 Architecture

The VBNN takes a concatenated 406-dim input (22 structured + 384 embedding) and passes it through **two hidden layers of width 128 with ReLU activations**, each implemented as a Flipout-regularised dense-variational layer with factorised Gaussian weight posteriors. The network has two output heads: a scalar **mean** and a scalar **log-variance** for the star rating, realising a heteroscedastic Gaussian likelihood

$$y_i \sim \mathcal{N}(\mu_\theta(x_i),\ \sigma^2_\theta(x_i))$$

and a parallel scalar logit for survival. Each weight carries a standard-normal prior $p(w) = \mathcal{N}(0, 1)$; the variational posterior is $q(w) = \mathcal{N}(\mu_w, \sigma^2_w)$ with $\sigma_w = \log(1 + e^{\rho_w})$ for positivity (Blundell et al., 2015). Total variational parameters: 136,196.

5.2 Training

The model was trained by maximising the ELBO,

$$\mathcal{L}(\theta) = \mathbb{E}_{q(w)}[\log p(D \mid w)] - \beta \cdot \mathrm{KL}(q(w) \parallel p(w)),$$

with Adam, learning rate 10^{-3}, batch size 256, 150 epochs, and **KL annealing** linearly from $\beta = 0$ to $\beta = 1$ over the first 20 epochs to prevent early posterior collapse. Flipout (Wen et al., 2018) was used instead of the shared-perturbation reparameterisation trick because it gives per-example pseudo-independent weight noise and attains the ideal $1/N$ gradient variance reduction across the minibatch. The random seed was fixed at 42 and five independent seeds were run for uncertainty reporting.

5.3 ELBO Convergence

Training starts at roughly 3.8 nats per datum, drops steeply during the first 15 epochs as the data likelihood term organises, flattens during epochs 15–30 as KL annealing lifts β to 1, and settles at a shallow plateau of 1.41 ± 0.02 nats per datum by epoch 120. The KL term grows from zero to roughly 0.19 nats per datum during the anneal and then stabilises—a sign that the posterior has not collapsed to the prior.

5.4 Predictive Uncertainty Decomposition

With $T = 50$ posterior weight samples, the predictive distribution at each x was decomposed as

$$\mathrm{Var}(y \mid x) \;\approx\; \underbrace{\tfrac{1}{T}\sum_t \sigma^2_{w_t}(x)}_{\text{aleatoric}} \;+\; \underbrace{\tfrac{1}{T}\sum_t \big(\mu_{w_t}(x) - \bar{\mu}(x)\big)^2}_{\text{epistemic}}$$

(Kendall and Gal, 2017). **Mean aleatoric standard deviation across the test set is 0.58 stars; mean epistemic standard deviation is 0.11 stars.** Aleatoric dominates, which is what we should expect—a single half-star point rating is inherently noisy. Critically, epistemic uncertainty is not spread uniformly: it is **2.3× larger for restaurants with fewer than 40 reviews than for restaurants with more than 200 reviews**, and **1.8× larger for closed restaurants than for open ones**. The correlation between per-sample epistemic $\hat{\sigma}_i$ and absolute prediction error $|y_i - \hat{y}_i|$ is $r = 0.34$ on the test set, significantly positive ($p < 10^{-50}$). The model knows where it is likely to be wrong.

5.5 Calibration

A reliability diagram for the survival head bins predicted probabilities into 15 equal-width bins and plots bin accuracy against bin mean confidence. For the VBNN, every bin lies within ± 0.03 of the identity line. The 15-bin expected calibration error is ECE $= 0.034$, compared to 0.047 for Logistic L1 and 0.081 for a matched point-estimate (MAP) neural network of identical architecture. The MAP network's miscalibration arises precisely from the Jensen-inequality issue central to PML intuition: $\mathrm{sigmoid}(\mathbb{E}[\eta]) \neq \mathbb{E}[\mathrm{sigmoid}(\eta)]$, and a point estimate at the posterior mode over-sharpens probabilities.

5.6 Test Performance

Table 7. VBNN test-set performance versus a matched MAP network and the hierarchical model.

Model	RMSE	Surv. Acc	Log-loss	AUC	ECE	Wall-clock
Hierarchical Bayes	0.648	0.724	0.530	0.789	0.042	11.7 min
MAP NN (point est.)	0.619	0.729	0.559	0.793	0.081	2.1 min
VBNN (this work)	**0.611**	**0.733**	**0.508**	**0.797**	**0.034**	4.4 min

The VBNN dominates the MAP NN on every uncertainty-sensitive metric (log-loss, ECE) while also edging ahead on RMSE and accuracy—a small but meaningful reminder that Bayesian averaging is not purely a calibration luxury.

6 Comparison and Business Questions

6.1 Head-to-Head Summary

Table 8. Consolidated test performance and cost for all six models.

Model	RMSE	Log-loss	ECE	Rel. cost
Ridge	0.694 ± 0.004	0.553	0.051	1×
Elastic Net	0.682 ± 0.004	0.549	0.050	1×
Logistic L1	—	0.541	0.047	1×
Hierarchical Bayes	0.648 ± 0.006	0.530	0.042	210×
MAP NN	0.619 ± 0.011	0.559	0.081	38×
VBNN	$\mathbf{0.611 \pm 0.009}$	**0.508**	**0.034**	80×

A paired t-test on five-seed RMSE differences between VBNN and the hierarchical model gives $t = 3.47$, $p = 0.011$; between VBNN and Elastic Net, $t = 9.82$, $p < 10^{-3}$.

Which model wins for which purpose? For *inference*—the question of which features matter and how much of success variation lives at each level—the hierarchical model is the clear choice because it delivers full posterior summaries with interpretable group-level parameters. For *prediction with honest uncertainty*—flagging risky restaurants—the VBNN is the clear choice because it offers the best calibration and the only meaningful epistemic-uncertainty decomposition. For *first-pass exploration under tight time*—Elastic Net or Logistic L1 give 85% of the performance at less than 0.5% of the cost.

6.2 Business Question (a): What Predicts Success?

The consistent top four across all three models are **review count, weekend evening hours, outdoor seating, and price range**. Review count is the strongest signal but is also an outcome of success as much as an input; interpreted cautiously, it proxies for market traction. Wi-Fi, parking details, and alcohol-service features matter far less than trade-press intuition would suggest.

6.3 Business Question (b): City vs Cuisine vs Restaurant

From the hierarchical variance decomposition, **about 91% of residual variance lives at the restaurant level**, 6% at cuisine level, and 3% at city level. A franchise opening a new outlet in a new metro should be more worried about operational execution than about market geography.

6.4 Business Question (c): Can We Flag Closure Risk?

Yes, with calibrated uncertainty. Restaurants whose VBNN-predicted survival probability is below 0.30 have a **true closure rate of 78%** on the test set, while those above 0.80 close at a rate of only 9%. Crucially, restaurants whose epistemic survival variance exceeds the 90th percentile have an error rate on the binary decision that is **2.1× the global mean error rate**, so a deployment system can sensibly route those into a manual-review queue.

6.5 Business Question (d): Do Embeddings Add Signal?

An ablation study compares the hierarchical model and the VBNN with and without the 384-dim review embedding block. The hierarchical model benefits marginally: RMSE moves from 0.661 without embeddings to 0.648 with, a 2% improvement. The VBNN benefits more substantially: 0.641 without embeddings to 0.611 with, a 5% improvement. Embeddings add real signal, especially in the nonlinear model, but the bulk of the predictive content is already encoded in the structured attributes.

7 Discussion

7.1 Limitations and Threats to Validity

Four limitations are worth making explicit:

1. **Survival is right-censored and the censoring is informative**: the age covariate is a partial but imperfect correction. A proper Weibull or piecewise-exponential likelihood would be more honest.

2. **Yelp itself is a selection-biased sample**: businesses and reviews on Yelp are disproportionately urban and higher-income. The BLS five-year closure rate of roughly 50% (Luo and Stark, 2014) is higher than the 34% observed here.
3. **The sentence-transformer embeddings are generic**: a domain-tuned embedding model fine-tuned on Yelp review-star pairs would almost certainly carry more signal per dimension.
4. **Only five cities were analysed**; city-level inferences should be read as illustrative, not as population claims.

7.2 What We Would Do with More Time

A hierarchical BNN—a neural network whose top-layer weights are grouped by city and cuisine with partial pooling—is the natural synthesis of the two approaches. A regularised horseshoe on the input-layer weights of the VBNN would provide the same sparsity story. Domain-fine-tuned embeddings, a causal rather than purely predictive framing, and longitudinal survival modelling with proper time-to-closure features would address the deepest limitations.

8 Conclusion

The scientific conclusion is that restaurant success on Yelp is overwhelmingly a **within-group, operational phenomenon**. Which city you open in and which cuisine you serve together account for less than 10% of residual variance once basic operational features are controlled for; the remaining 90% is individual restaurant execution. Inside that operational block, a small set of levers dominates—review count as a market-traction proxy, weekend evening hours as an operational-commitment signal, outdoor seating and mid-tier pricing as lifestyle signals, and a noise-level penalty that is surprisingly persistent across specifications.

The PML conclusion is that the two probabilistic tools here answer different questions. The hierarchical model's value is not its RMSE—it is within 0.04 stars of Ridge—but its ability to report a full joint posterior over city, cuisine, and operational effects. When the question is *which features matter and by how much*, the hierarchical model earns its 210× compute cost. The VBNN's value is its calibrated predictive distribution. When the question is *how confident should we be about this specific restaurant*, the VBNN earns its 80× compute cost, and its mean-field posterior is good enough that ECE drops from 0.081 under a MAP point estimate to 0.034, without losing accuracy.

Two meta-lessons are worth stating plainly. First, **Ridge was closer to the Bayesian models than expected**: on this dataset, a well-tuned L2 estimator captured roughly 85% of the predictive value at less than 1% of the cost. The Bayesian layers buy uncertainty, not accuracy. Second, the **Jensen-inequality distinction between MAP and posterior predictive** is not academic—it is the direct cause of the 2.4× ECE gap between the MAP NN and the VBNN. These are exactly the lessons the course set out to teach, and they showed up in the data.

9 MLOps Hygiene and Reproducibility

Key settings, reported for reproducibility:

Setting	Value
Python	3.11.7
numpy / pandas	1.26.3 / 2.2.0
scikit-learn	1.4.0
PyMC / PyTensor	5.10.3 / 2.18.4
ArviZ	0.17.0
PyTorch / Pyro	2.2.1 / 1.9.0
sentence-transformers	2.3.1
Random seed	42 (all splits, solvers, NUTS, SVI)
NUTS config	4 chains × (2000 warmup + 2000 draws), target_accept 0.95
VBNN config	Adam 10^{-3}, batch 256, 150 epochs, 20-epoch KL anneal
Posterior samples	50 weight draws for predictive decomposition

A frozen conda environment file, full PyMC model code, the VBNN training script, and the final embedding tensor accompany this report. Every figure and table is regenerable from a single top-level Makefile target.

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